MTA Daily Ridership: Post-COVID Recovery Analysis & Next-Month Forecast

DEPI – Data Analysis Track | Phase 1: Planning & Requirements | 16 October 2026

Team: Salem Emad, Mohamed Hossam, Dalia El-Barbary, Rana Osama, Mira Morris

KPIs Document

1. Purpose

This document defines how the success of the project is measured. KPIs are split into two types: Project KPIs, which measure the success of the work itself, and Analysis KPIs, which are the numbers that will appear in the Power BI dashboard. In Phase 1 each KPI is a written definition only; in Phase 3 the same KPIs become Power BI measures and dashboard cards.

2. Project KPIs

KPI	Definition	Formula	Target	Data source	When measured
On-time delivery	Share of the 4 phases submitted by their official deadline	Phases delivered on time / 4 × 100	100%	Gantt chart (02_Gantt_Tasks_Risks.xlsx)	At each deadline (16 Oct, 6 Nov, 30 Nov, 4 Dec)
Forecast accuracy (MAPE)	Average daily percentage error of the total passenger ridership forecast, measured by a backtest	average(|Actual − Forecast| / Actual) × 100, over the daily values of the last 3 months (Aug–Oct 2024), with the model trained only on data before August 2024	≤ 10%	Total Passenger Ridership (actual) vs. forecast (same weekday last year × recent growth)	Phase 3, after the model is built
Dashboard completeness	Share of planned dashboard pages that are finished	Pages completed / Pages planned × 100	100% (at least 4 pages)	Power BI file	Weekly during Phase 3
Team contribution	Every member contributes visibly to the repository	Number of commits per member per week	≥ 1 commit per member per week	GitHub commit history	Weekly
Requirements coverage	Share of user stories linked to at least one functional requirement	Stories linked to ≥ 1 FR / Total stories × 100	100%	04_Requirements.pdf (traceability table)	At Phase 1 review

3. Analysis KPIs

KPI	Definition	Formula	Target	Data source	When measured
Total Passenger Ridership (and per-mode totals)	Sum of daily ridership for the selected mode and period; for the five passenger modes combined it is Total Passenger Ridership	SUM of the mode's "Total Estimated Ridership" column	Shown per mode; the five passenger modes also shown as one Total Passenger Ridership	MTA_Daily_Ridership.csv	Phase 3 (dashboard card)

KPI	Definition	Formula	Target	Data source	When measured
% Recovery vs. pre-pandemic	Average of the "% of Comparable Pre-Pandemic Day" column for the selected period	AVERAGE of the mode's "% of Comparable Pre-Pandemic Day" column	Shown per mode against a 100% reference line	MTA_Daily_Ridership.csv	Phase 3
Overall Passenger Recovery %	Recovery of the five passenger modes combined, weighted by ridership (the Overall Recovery % scorecard card)	(SUM of ridership / SUM of implied baseline) × 100, where implied baseline = ridership ÷ (% / 100); rows with null or zero % are excluded	Shown as a scorecard card and by period	MTA_Daily_Ridership.csv	Phase 3 (dashboard card)
YoY Growth	Change in ridership compared with the same period last year	(This year − Same period last year) / Same period last year × 100	Shown per mode	MTA_Daily_Ridership.csv	Phase 3
Weekday vs. Weekend ratio	How much busier weekdays are than weekends	Avg weekday ridership / Avg weekend ridership (Sat–Sun)	Shown per mode	MTA_Daily_Ridership.csv	Phase 3
Busiest weekday / month	The weekday and the month with the highest average ridership	Day or month with the MAX of average ridership	Shown per mode	MTA_Daily_Ridership.csv	Phase 3
Top 3 peak days / lowest months	The three days with the highest total passenger ridership and the calendar months with the lowest average ridership	TOP 3 days by passenger ridership; month average = AVERAGE of daily ridership per calendar month, lowest = MIN	Top 3 days for the passenger total; lowest months shown per mode	MTA_Daily_Ridership.csv	Phase 3
Next-month forecast	Predicted total passenger ridership for November 2024	Forecast model output, summed over the 30 days of November 2024	Error ≤ 10% (MAPE)	Forecast model	Phase 3

4. Notes and Data Limitations

- Passenger modes (Subways, Buses, LIRR, Metro-North, Staten Island Railway) may be totalled together. Access-A-Ride (scheduled trips) and Bridges & Tunnels (vehicle traffic) are reported separately because their units differ.
- Pre-pandemic percentages come from the source data. In February 2023 the MTA upgraded its method and refreshed all percentages in the dataset, so the whole series uses the same method: each day is compared with the 2019 monthly average for the same day type (non-holiday weekday, Saturday, or Sunday/holiday) in the equivalent month.
- Staten Island Railway contains zero values in the daily ridership column (32 days) and in the pre-pandemic percentage column (54 days). The source data has no blank cells, so these zeros will be reviewed during data cleaning and treated as probable missing observations (converted to nulls where confirmed). The decision will be documented.
- YoY Growth is available from March 2021 onward, and 2024 is compared with the same months (January–October) of the previous year.
- A backtest on August–October 2024 gave a daily MAPE of 6.87%, below the 10% target. The model was trained only on data before August 2024 and uses the same weekday of the previous year, scaled by the growth of the last 8 weeks. The error grows with distance from the training period (about 3% in August and about 9% in October), and moving holidays such as Thanksgiving can reduce accuracy. The final forecast model will be built and tested in Phase 3.
- Bus figures from 23 Mar to 30 Sep 2020, and LIRR and Metro-North figures, are model-based estimates, not exact counts.

5. How We Measure

The team reviews the Project KPIs weekly (deadlines, GitHub commits, dashboard pages). The Analysis KPIs are calculated in Power BI in Phase 3 and checked against the source data. The targets are linked to the Proposal objectives: the forecast accuracy target (MAPE of 10% or less) and the dashboard target (at least 4 pages).